

03_Laboratory_Exercise_1 - Active Attack vs Passive Attack

a. Summary of the article:

The article explains the difference between active and passive attacks in cybersecurity. It emphasizes that the internet is full of risks, and modern attackers are more sophisticated than before. Passive attacks involve monitoring systems and networks without altering data, making them very hard to detect. Examples include scanning for open ports, expired certificates, and footprinting to collect intelligence for later use. Active attacks, on the other hand, involve modifying or exploiting the gathered information to compromise users or networks. These attacks can disrupt data integrity and availability, making them more noticeable but still dangerous. The article highlights that both types of attacks can cause serious harm, and individuals and organizations must adopt proactive measures to protect their data and systems.

b. Compare and contrast active and passive attacks:

Passive Attack: Focuses on monitoring and collecting information without altering the system. It is difficult to detect since no changes are made to data or processes. Its main goal is intelligence gathering.

Active Attack: Uses the information obtained (often from passive attacks) to compromise a system by altering, blocking, or misusing data. It is easier to detect because it disrupts normal operations, but identifying the root cause is challenging.

Contrast: Passive attacks are about observation and stealth, while active attacks are about manipulation and direct interference.

c. Recommendations to prevent these attacks:

- Use strong encryption to protect data from being read during transmission.
- Regularly update and patch systems to close vulnerabilities.
- Avoid posting sensitive personal or company information publicly.
- Implement strong authentication and identity protection measures.
- Educate employees and individuals to recognize risks and practice safe online habits.
- Use proper monitoring tools to detect unusual network activity early.
- Adopt a mindset that everyone online is a potential target and security must be a shared responsibility.